

Demonstration of Proposed Features

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation	Predicts the most suitable crop based on soil nutrients and environmental conditions using Machine Learning.	Implemented	Yes	Core functionality successfully demonstrated.
2	User-Friendly Web Interface	Responsive web interface with Home, About, and Prediction pages built using HTML, CSS, and Flask.	Implemented	Yes	Easy navigation and professional design.
3	Machine Learning Integration	Trained Scikit-learn model integrated with Flask backend for real-time prediction.	Implemented	Yes	Model successfully loads and predicts crops.
4	Input Validation	Accepts valid numerical inputs for soil and weather parameters with placeholders and required fields.	Implemented	Yes	Prevents invalid or incomplete input.
5	Prediction Result Display	Displays the recommended crop clearly after processing user input.	Implemented	Yes	Includes visually enhanced result section
6	Future Enhancements	Fertilizer recommendation, weather API integration, disease detection, multilingual support.	Pending	No	Planned for future project versions.

Feature Implementation Summary:

Total Features Proposed	5
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	100%